

Team Number: Group 1

Project Title: Syria FireWatch

Climate Challenge: Climate Risks and Natural Hazards (Wildfire)

Problem Statement:

Wildfire risk along the Syrian coast may increase during periods of high temperature, low humidity, strong winds, and low rainfall. However, the relationship between these weather conditions and fire activity has not been clearly measured. This project will use satellite fire detections and daily weather data from the Latakia and Tartus region to identify the weather conditions associated with fire events and examine changes in wildfire activity over time.

Main Research Question:

Which weather conditions are most associated with satellite-detected fire days, and have high-risk wildfire days increased over time?

Dataset Name and Source/Link:

- **NASA FIRMS Active Fire Detections (VIIRS/MODIS)** — <https://firms.modaps.eosdis.nasa.gov>
- **NASA POWER Daily Weather Data** — <https://power.larc.nasa.gov>
Variables: maximum temperature (T2M_MAX), relative humidity (RH2M), wind speed (WS2M), and precipitation (PRECTOTCORR).

Expected Project Output:

Python analysis and visualizations showing wildfire trends, the relationship between weather conditions and fire events, and a classification of days into low, moderate, and high wildfire-risk levels.